Reward, Not Residuals: Model-Based Actor-Critic Learning Recovers the Analytic Climate Economy

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Abstract

Deep learning methods for dynamic economic models typically train a network to satisfy hand-derived equilibrium conditions, which converge to the wrong solution if a condition is derived incorrectly. I ask whether a reinforcement-learning method that uses only the reward and the laws of motion can instead recover the exact solution of a climate-economy model. In the Analytic Climate Economy with DICE 2016 and DICE 2023 functional forms, a model-based actor-critic that differentiates through the known dynamics recovers the optimal consumption and abatement paths, with a mean abatement error between 0.010 and 0.016 across six training runs, and the social cost of carbon implied by the gradient of its critic is within 1–4% of the closed-form value. The welfare loss relative to the optimum is below 0.04% of consumption. A critic trained only on values, however, fits the value function closely but implies a social cost of carbon with the wrong sign, and the policy stops abating. The social cost of carbon is a small gradient of a large function, and recovering it requires training the critic on the gradient of the Bellman equation as well (value-gradient learning), with each component normalized by its own scale. A correctly specified equilibrium-condition method remains about an order of magnitude more precise, but the reward-based method needs no derived conditions.

1 Introduction

Deep learning methods for dynamic economic models fall, broadly, into two families. The first trains a neural network to satisfy the model's equilibrium conditions: Euler equations, first-order conditions, envelope conditions and complementary slackness conditions are written as residuals that vanish at the solution, and the network minimizes their squared sum on simulated states. Deep Equilibrium Nets (DEQNs) (Azinovic et al. 2022) are the leading example, and they have been applied to integrated assessment models (IAMs) of climate and the economy (Folini et al. 2024; Friedl et al. 2023). The second family, reinforcement learning (RL), maximizes the discounted sum of rewards directly and never writes down an optimality condition (Sutton 2018; Bertsekas 2019).

The two families are often presented as close relatives, and they share the structure of a Markov decision process. They differ, however, in what is minimized. A residual-based loss is a consistency criterion: it is zero wherever the hand-derived conditions hold, and it is only as correct as the derivation behind it. If an envelope condition omits a term, the network converges to a wrong fixed point without any sign of failure, since the loss still goes to zero. A reward-based objective, in contrast, is defined by the economic primitives alone (utility, technology and the laws of motion), so there is nothing to derive and nothing to get wrong beyond the model itself.

This paper asks whether a reward-based method can match the accuracy of a correctly specified

DEQN on a problem where the exact answer is known. I use the Analytic Climate Economy (ACE) of Traeger (2023) with the production functions of DICE 2016 and DICE 2023. ACE has a closed-form solution for the consumption rate, the abatement rate, the costates and the social cost of carbon (SCC), which provides an exact benchmark. The motivating example is close at hand: in an earlier version of a DEQN for this model, errors in the envelope condition for capital and double-counted terms in the first-order conditions led the network to a consumption rate between 0.53 and 0.93 instead of the optimal 0.74, and to an abatement rate near zero.

The method is a model-based actor-critic. The critic learns the value of the current policy from the Bellman equation, and the actor maximizes the one-step lookahead, current utility plus the discounted critic value of the next state, by differentiating through the known laws of motion. This is the deterministic policy gradient (Silver et al. 2014) with a known, differentiable model in place of a learned action-value function, as in stochastic value gradients (Heess et al. 2015). Nothing in the objective is derived by hand.

The main finding is that a plain version of this method fails, and that the reason is economically interesting. With the critic trained by temporal-difference learning alone, the actor learns a reasonable consumption rate but abandons abatement entirely, because the critic attributes part of the growth in value over time to the rising carbon stock and so estimates a *positive* marginal value of atmospheric carbon. The damages from one more gigaton of carbon change the value function by about 10^{-4}, far below the precision of a value function fitted to values of order 50. The SCC is a small gradient of a large function, and the policy depends on that gradient, not on the value itself.

Training the critic on its gradient as well as its value resolves the problem. The target gradient is the gradient of the right-hand side of the Bellman equation with respect to the state, computed by automatic differentiation through the known dynamics. This is value-gradient learning (Werbos 1992; Fairbank and Alonso 2012), a form of Sobolev training (Czarnecki et al. 2017), and it is the reward-based counterpart of the envelope conditions in a DEQN, except that the conditions are generated by the computer rather than derived by the researcher. With each gradient component normalized by a running estimate of its own scale, the actor-critic recovers the analytic consumption rate, abatement path, costates and SCC in both versions of the model. Across three random seeds per model, its SCC is within 1–4% of the closed-form value and its welfare loss relative to the optimum is below 0.04% of consumption. A DEQN with correctly derived conditions remains about an order of magnitude more precise, which is the price of learning shadow prices from the curvature of a value function rather than from conditions supplied by the researcher.

The rest of the paper is organized as follows. Section 2 relates the approach to the literature. Section 3 summarizes the model and its analytic solution. Section 4 presents the actor-critic method, and Section 5 the experimental setup. Section 6 reports the results, Section 7 discusses when reward-based and residual-based methods are each appropriate, and Section 8 concludes.

2 Related Literature

Residual-based deep learning in economics. Minimizing the residuals of equilibrium conditions has a long history in economics through projection and collocation methods (Judd 1992; Judd et al. 2011), which represent the policy with polynomials. DEQNs (Azinovic et al. 2022) and related deep-learning methods (Maliar et al. 2021; Kahou et al. 2021) replace the polynomials with neural networks and train them on simulated states, which mitigates the curse of dimensionality. Folini

et al. (2024) and Friedl et al. (2023) apply DEQNs to climate-economy models. Outside economics, the same idea underlies physics-informed neural networks (Raissi et al. 2019), which minimize the residuals of differential equations. A strength of residual-based methods is that they apply to equilibria that are not the solution of any single optimization problem, such as decentralized economies with market clearing.

Reinforcement learning and optimal control. Dynamic programming and RL share the Bellman equation as their foundation (Bertsekas 2019; Sutton 2018). Deterministic policy gradients (Silver et al. 2014) and their deep variants (Lillicrap et al. 2016; Fujimoto et al. 2018) train a deterministic actor on the gradient of a learned critic. When the dynamics are known and differentiable, the critic of the next state can be evaluated through the model, as in stochastic value gradients (Heess et al. 2015). The delayed actor updates and target networks used here follow Fujimoto et al. (2018) and Lillicrap et al. (2016). RL has also been applied to climate-economy models (Strnad et al. 2019; Wolf et al. 2023), but, to my knowledge, not evaluated against a closed-form solution of a quantitative IAM.

Value-gradient learning. Dual heuristic programming (Werbos 1992) and value-gradient learning (Fairbank and Alonso 2012) train a critic on the gradient of the value function rather than, or in addition to, its level. Sobolev training (Czarnecki et al. 2017) shows more generally that fitting derivatives improves the accuracy and data efficiency of neural approximations. In economics, the gradient of the value function is the vector of costates or shadow prices, which makes these methods a natural fit for problems in which policy depends on shadow prices, such as the SCC.

3 The Analytic Climate Economy

I use ACE (Traeger 2023) with the DICE production function, carbon cycle and two-layer temperature model, in the calibrations of DICE 2016 (Nordhaus 2018) and DICE 2023 (Nordhaus and Barrage 2023), following Traeger (2021). Time is discrete with periods of $\Delta = 5$ years.

States, controls and reward. The state is $\boldsymbol{s}_t = (k_t, \boldsymbol{M}_t, \boldsymbol{\tau}_t, t)$, where $k_t = \log K_t$ is log capital, $\boldsymbol{M}_t \in \mathbb{R}^3$ are the carbon stocks in the atmosphere, upper ocean and deep ocean (GtC), and $\boldsymbol{\tau}_t \in \mathbb{R}^2$ are transformed temperatures, $\tau_{i,t} = \exp(\xi_1 T_{i,t})$. The planner chooses the consumption rate $x_t \in (0,1)$ and the abatement rate $\mu_t \in [0,1]$. Output after abatement costs is

$$\log F_t = \log A_t + \kappa k_t + (1-\kappa)\log N_t + \log\left(1 - \theta_{1,t}\mu_t^{\theta_2}\right), \tag{1}$$

damages are $1 - D(T_{1,t}) = \exp(-\xi_0\tau_{1,t} + \xi_0)$, and the reward is log consumption,

$$u(\boldsymbol{s}_t, \boldsymbol{a}_t) = \log x_t + \log F_t - \xi_0\tau_{1,t} + \xi_0. \tag{2}$$

Laws of motion. Capital, carbon and temperatures evolve as

$$k_{t+1} = \log F_t - \xi_0\tau_{1,t} + \xi_0 + \log(1-x_t) + \log\frac{1+g_k\Delta}{\delta_k + g_k}, \tag{3}$$

$$\boldsymbol{M}_{t+1} = \boldsymbol{\Phi}\boldsymbol{M}_t + \boldsymbol{e}_1\zeta\left[(1-\mu_t)E_t^{\mathrm{BAU}} + E_t^{\mathrm{land}}\right], \tag{4}$$

$$\boldsymbol{\tau}_{t+1} = \boldsymbol{\sigma}\boldsymbol{\tau}_t + \boldsymbol{e}_1\sigma^{\mathrm{forc}}\frac{M_{1,t}}{M_{\mathrm{pre}}}, \tag{5}$$

where emissions are measured in $GtCO_2$ per year and $\zeta = \Delta/3.664$ converts them into GtC per period. The planner maximizes $\sum_t \beta^t u(\boldsymbol{s}_t, \boldsymbol{a}_t)$ with $\beta = (1+\rho)^{-\Delta}$.

Business-as-usual emissions. In DICE, business-as-usual emissions are proportional to gross output, $E_t^{\text{BAU}} = \sigma_t Y_t^{\text{gross}}$, and therefore depend on capital. The closed-form solution of ACE requires that the planner takes E_t^{BAU} as given. For an exact benchmark, I fix E_t^{BAU} to an exogenous reference path: $\sigma_t Y_t^{\text{gross}}$ evaluated along the analytic optimal path. On that path the reference coincides with the DICE definition; off the path it remains exogenous, so the problem solved by the learning method and the problem with the closed-form solution are the same.

Analytic solution. Because the value function is affine in $(k_t, \boldsymbol{M}_t, \boldsymbol{\tau}_t)$, the costates $\boldsymbol{\lambda}_t = \beta \nabla V(\boldsymbol{s}_{t+1})$ are constant over time:

$$\lambda_k = \frac{\beta\kappa}{1-\beta\kappa}, \tag{6}$$

$$\boldsymbol{\lambda}_\tau = -\beta\xi_0(1+\lambda_k)\left(\boldsymbol{I} - \beta\boldsymbol{\sigma}^\top\right)^{-1}\boldsymbol{e}_1, \tag{7}$$

$$\boldsymbol{\lambda}_M = \beta\frac{\sigma^{\text{forc}}}{M_{\text{pre}}}\lambda_{\tau_1}\left(\boldsymbol{I} - \beta\boldsymbol{\Phi}^\top\right)^{-1}\boldsymbol{e}_1. \tag{8}$$

The optimal consumption rate is $x^* = 1 - \beta\kappa \approx 0.7395$. The optimal abatement rate equates the marginal product of emissions to the carbon price,

$$\frac{\theta_{1,t}\theta_2\mu_t^{\theta_2-1}}{E_t^{\text{BAU}}\left(1-\theta_{1,t}\mu_t^{\theta_2}\right)} = -\frac{\zeta\lambda_{M_1}}{1+\lambda_k}, \tag{9}$$

with $\mu_t^* = 1$ where the solution exceeds one, and the SCC in USD per tCO_2 is

$$SCC_t = -1000\, Y_t^{\text{net}}\,\frac{\zeta\lambda_{M_1}}{1+\lambda_k}. \tag{10}$$

I verify numerically that these costates equal β times the finite-difference gradient of the value of the analytic policy, and that perturbing the analytic policy lowers welfare.

4 Method

4.1 Model-Based Actor-Critic

The actor $\pi_\theta(\boldsymbol{s}) = (x, \mu)$ and the critic $V_\phi(\boldsymbol{s})$ are feed-forward networks with two hidden layers of 512 ReLU units. The actor's two outputs pass through a sigmoid, so that $x, \mu \in (0,1)$. The critic outputs $V_\phi = 40 + 30\, h_\phi(\boldsymbol{s})$, a fixed affine map of the network output h_ϕ onto the typical range of the value function. Both networks receive log capital, the carbon stocks in thousands of GtC, the transformed temperatures and the transformed time $1 - e^{-0.02t}$.

Write $f(\boldsymbol{s}, \boldsymbol{a})$ for the laws of motion. The critic is trained on the Bellman equation of the current policy,

$$\mathcal{L}_{\text{TD}}(\phi) = \mathbb{E}_{\boldsymbol{s}}\left[\left(V_\phi(\boldsymbol{s}) - y(\boldsymbol{s})\right)^2\right], \qquad y(\boldsymbol{s}) = u(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) + \beta V_{\bar{\phi}}\left(f(\boldsymbol{s}, \pi_\theta(\boldsymbol{s}))\right), \tag{11}$$

where $V_{\bar{\phi}}$ is a target critic whose weights follow ϕ by Polyak averaging, $\bar{\phi} \leftarrow 0.99\bar{\phi} + 0.01\phi$. The actor maximizes the one-step lookahead

$$J(\theta) = \mathbb{E}_{\boldsymbol{s}} \left[u(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) + \beta V_\phi \left(f(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) \right) \right], \tag{12}$$

by gradient ascent, differentiating through u and f. Since the dynamics are deterministic and known, this gradient is exact given the critic. The objective contains only the reward and the laws of motion: there are no first-order conditions, envelope conditions or costate outputs.

4.2 Value-Gradient Learning

The actor's choice of abatement depends on the critic only through $\partial V_\phi / \partial M_1$, the marginal value of atmospheric carbon. At the solution, this derivative is about $-1.2 \cdot 10^{-4}$ per GtC, so that a change of 100 GtC moves the value by about 0.012, while the value itself is of order 50. A critic fitted only to values has no reason to get this derivative right, and in practice it does not (Section 6.5).

I therefore also train the critic on its gradient. Differentiating the Bellman equation with respect to the state gives a target for the gradient of the value function,

$$\boldsymbol{g}(\boldsymbol{s}) = \nabla_{\boldsymbol{s}} \left[u(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) + \beta V_{\bar{\phi}} \left(f(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) \right) \right], \tag{13}$$

computed by automatic differentiation, including the dependence of the policy on the state. The critic loss becomes

$$\mathcal{L}(\phi) = \mathcal{L}_{\mathrm{TD}}(\phi) + \mathbb{E}_{\boldsymbol{s}} \left[\sum_{i=1}^{6} \left(\frac{\partial_i V_\phi(\boldsymbol{s}) - g_i(\boldsymbol{s})}{\hat{s}_i} \right)^2 \right], \tag{14}$$

where $\hat{s}_i$ is the root of an exponential moving average of g_i^2 over mini-batches. This normalization is generic: it gives each component of the gradient the same weight regardless of its units, without economic knowledge of the magnitude of the costates. For a residual-based method, the analogous scaling must be chosen by hand.

The gradient target is the reward-based counterpart of the envelope conditions in a DEQN. The difference is who derives it: here, automatic differentiation of the Bellman equation produces the conditions, including every term that a hand derivation might omit.

4.3 State Distribution

The critic is trained on the states visited by the current policy and on perturbed copies of them, in which log capital is shifted by $\mathcal{N}(0, 0.2^2)$, the carbon stocks are scaled by $1 + \mathcal{N}(0, 0.15^2)$ and the transformed temperatures by $1 + \mathcal{N}(0, 0.05^2)$. Without perturbations, the states along a single deterministic trajectory move together over time, and the gradient of the critic in directions away from the trajectory is not identified.

4.4 Training Schedule

Each iteration simulates one episode of 128 periods with the current policy, adds three perturbed copies of each state, and runs two epochs of mini-batch updates with batches of 128 states. The actor is updated on every second critic update (Fujimoto et al. 2018). During the first 500 iterations,

only the critic is trained, so that the actor does not follow an uninformed critic to extreme controls, where the sigmoid saturates and gradients vanish. Both networks use Adam (Kingma and Ba 2017) with cosine learning-rate decay to 1% of the initial rate (10^{-3} for the critic, 10^{-4} for the actor) and gradient-norm clipping at 1. Training runs for 4000 iterations. Algorithm 1 summarizes the method.

Algorithm 1 Model-based actor-critic with value-gradient learning

for iteration $= 1, \dots, 4000$ **do**
  Simulate $\boldsymbol{s}_0, \dots, \boldsymbol{s}_{127}$ with π_θ; add three perturbed copies of each state
  for each mini-batch B, two epochs **do**
    $y \leftarrow u(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) + \beta V_{\bar{\phi}}(f(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})))$ and $\boldsymbol{g} \leftarrow \nabla_{\boldsymbol{s}} y$ for $\boldsymbol{s} \in B$
    Update ϕ on $\mathcal{L}_{\mathrm{TD}} + \sum_i ((\partial_i V_\phi - g_i)/\hat{s}_i)^2$; update $\hat{s}$; $\bar{\phi} \leftarrow 0.99\bar{\phi} + 0.01\phi$
    if past warm-up and on every second step **then**
      Update θ by gradient ascent on $u(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})) + \beta V_\phi(f(\boldsymbol{s}, \pi_\theta(\boldsymbol{s})))$
    end if
  end for
end for

4.5 Recovering Costates and the SCC

After training, the costates follow from the critic by automatic differentiation, $\hat{\boldsymbol{\lambda}}_t = \beta \nabla V_\phi(\boldsymbol{s}_{t+1})$, and the SCC from equation (10). No costate is an output of any network, so these are genuine out-of-objective tests: the critic is trained on values and on its own consistency, never on the analytic costates.

5 Experimental Setup

I train one actor-critic for each of the DICE 2016 and DICE 2023 calibrations of ACE, with identical hyperparameters. The parameters of the economy and the climate system follow the respective DICE versions as implemented by Traeger (2021), with a pure rate of time preference of 2.86% per year and a capital share of $\kappa = 0.3$.

Evaluation. After training, I simulate the learned policy from the initial state for 128 periods and compare it with the analytic solution over the periods up to 2300. I report the deviation of the consumption rate from x^*, the deviation of the abatement rate from μ_t^* at the same date, the relative error of the costates $\beta \nabla V_\phi(\boldsymbol{s}_{t+1})$ and of the implied SCC, and two measures of the critic and the policy as a whole. The first is the relative error of the critic against the true value of the learned policy, computed by simulating the policy for 250 periods from each state. The second is the welfare loss of the learned policy relative to the optimum, as the permanent percentage reduction in consumption that makes the planner indifferent: $\exp\left((1-\beta)(V^*(\boldsymbol{s}_0) - V^\pi(\boldsymbol{s}_0))\right) - 1$. The weights used for evaluation are those after the final iteration; no checkpoint is selected using the benchmark.

Comparison with a DEQN. As a reference, I compare with a DEQN trained on correctly derived equilibrium conditions for the same model, with a 2×1024 network and hand-chosen scales

for the costate outputs and residuals. That DEQN treats E_t^{BAU} as given in its first-order conditions but evaluates it at the simulated capital stock, which differs slightly from the exogenous reference path used here. I therefore compare the two methods by their errors relative to their own analytic benchmark rather than by their paths.

Ablations. To identify which components matter, I retrain the DICE 2016 model (i) without the value-gradient term, so that the critic is trained by temporal-difference learning alone, and (ii) without state perturbations, so that the critic sees only the states along the simulated trajectory.

Computation. Each training run takes about 25 minutes on four CPU cores. Code, configurations, trained weights and training logs are available in the repository.

6 Results

6.1 Policies

Figure 1 compares the learned controls with the analytic solution along the trajectory simulated with the learned policy. The abatement rate follows the analytic path, which rises from about 9% in the first period to full abatement in 2175 (DICE 2016) and 2195 (DICE 2023), with a mean absolute deviation of 0.014 and 0.012 over the periods up to 2300 (Table 1). The learned policy starts slightly low, at 6% instead of 9%. The largest deviations, of about 0.06, occur at the kink where the analytic rate reaches one and the constraint $\mu \leq 1$ starts to bind; the sigmoid output of the actor rounds this kink off. The consumption rate stays within 0.013 of $x^* = 0.7395$ in the DICE 2016 model and within 0.007 in the DICE 2023 model. The actor has learned that the optimal consumption rate does not depend on the state, although it retains small, slow drifts over time.

	ACE-DICE 2016		ACE-DICE 2023	
	Actor-critic	DEQN	Actor-critic	DEQN
$\max_t \lvert x_t - x^* \rvert$	0.0126	0.0006	0.0069	0.0028
mean $\lvert \mu_t - \mu_t^* \rvert$	0.0141	0.0009	0.0123	0.0020
$\max_t \lvert \mu_t - \mu_t^* \rvert$	0.0582	0.0166	0.0602	0.0227
SCC, mean relative error	1.19%	0.18%	2.99%	0.56%
λ_k, mean relative error	4.88%	–	2.66%	–
λ_{M_1}, mean relative error	0.72%	–	2.54%	–
λ_{τ_1}, mean relative error	1.42%	–	1.91%	–
Welfare loss (consumption equivalent)	0.0388%	–	0.0039%	–

Table 1: Accuracy of the actor-critic and of a DEQN with correctly derived conditions, each against its analytic benchmark, over the periods up to 2300. Relative errors of the costates and the SCC are averaged over the same periods. The DEQN outputs costates directly, so its costate rows are reported in the companion thesis; its welfare loss is not computed.

Because the learned controls are close to optimal, the simulated paths of temperature, atmospheric carbon and capital are close to the analytic optimal path (Figure 2). Temperature peaks at 3.33°C

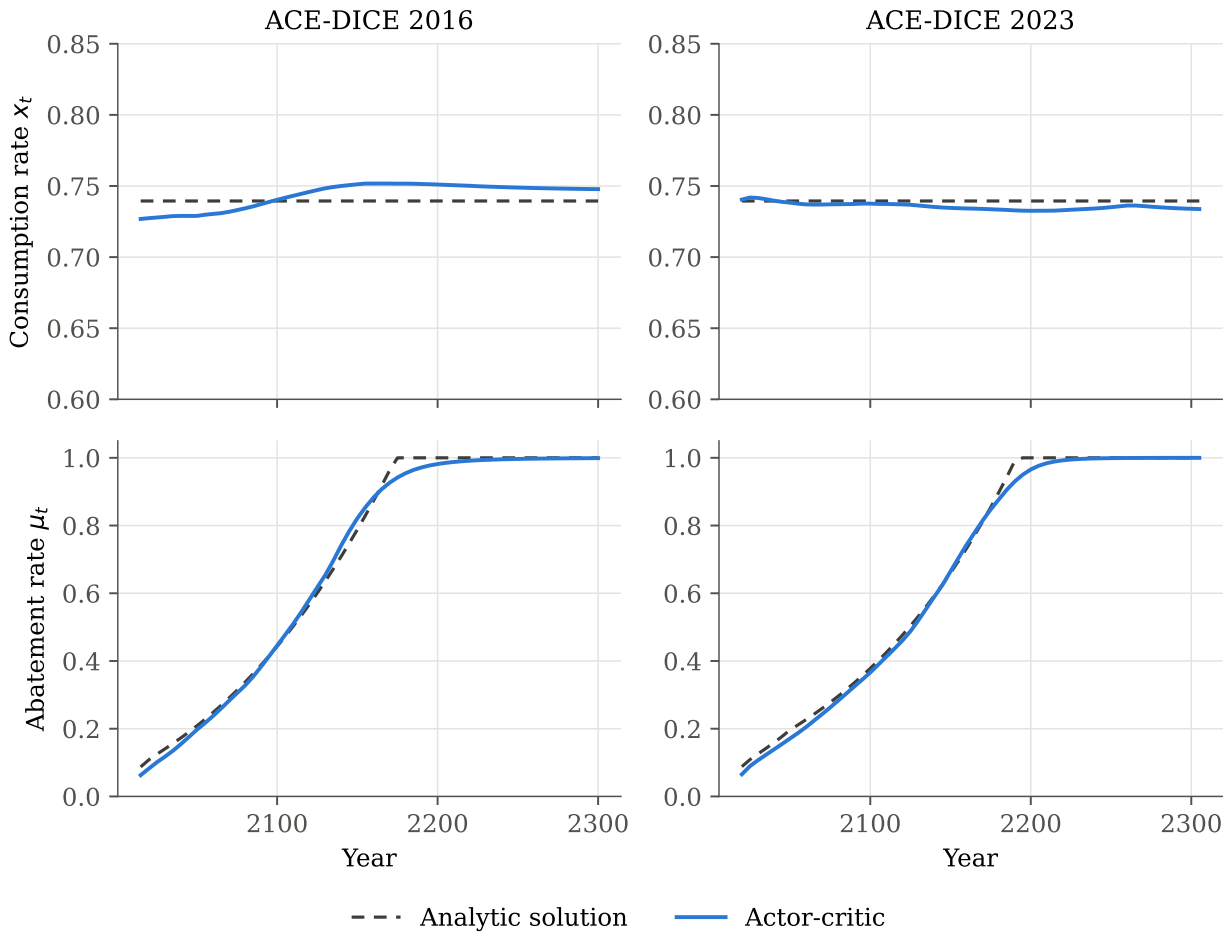


Figure 1: Consumption rate (top) and abatement rate (bottom) of the learned policy and of the analytic solution, along the trajectory simulated with the learned policy.

in 2145 in the DICE 2016 model, as on the analytic path, and at 4.75°C in 2190 in the DICE 2023 model, against 4.72°C on the analytic path. The visible deviations after 2200 in the DICE 2023 model follow from the slower approach to full abatement around the kink.

The welfare loss of the learned policy relative to the optimum is 0.039% of consumption in every period for the DICE 2016 model and 0.004% for the DICE 2023 model. These losses are small because the value function is flat around the optimum: deviations in the controls cause losses in welfare that are only of second order.

6.2 Costates and the Social Cost of Carbon

No network in the actor-critic outputs a costate. Nevertheless, $\beta \nabla V_\phi(\boldsymbol{s}_{t+1})$ recovers the analytic costates with mean relative errors between 0.7% and 4.9% (Figure 3). The carbon costate $\lambda_{M_1} \approx -1.05 \cdot 10^{-4}$, which is four orders of magnitude smaller than the capital costate and which determines abatement, is recovered to within 0.7% in the DICE 2016 model and 2.5% in the DICE 2023 model. The implied SCC follows the analytic SCC closely (Figure 4), with mean relative errors of 1.2% and 3.0%. In 2050, the actor-critic implies an SCC of 37.0 USD/tCO$_2$ in the DICE 2016 model

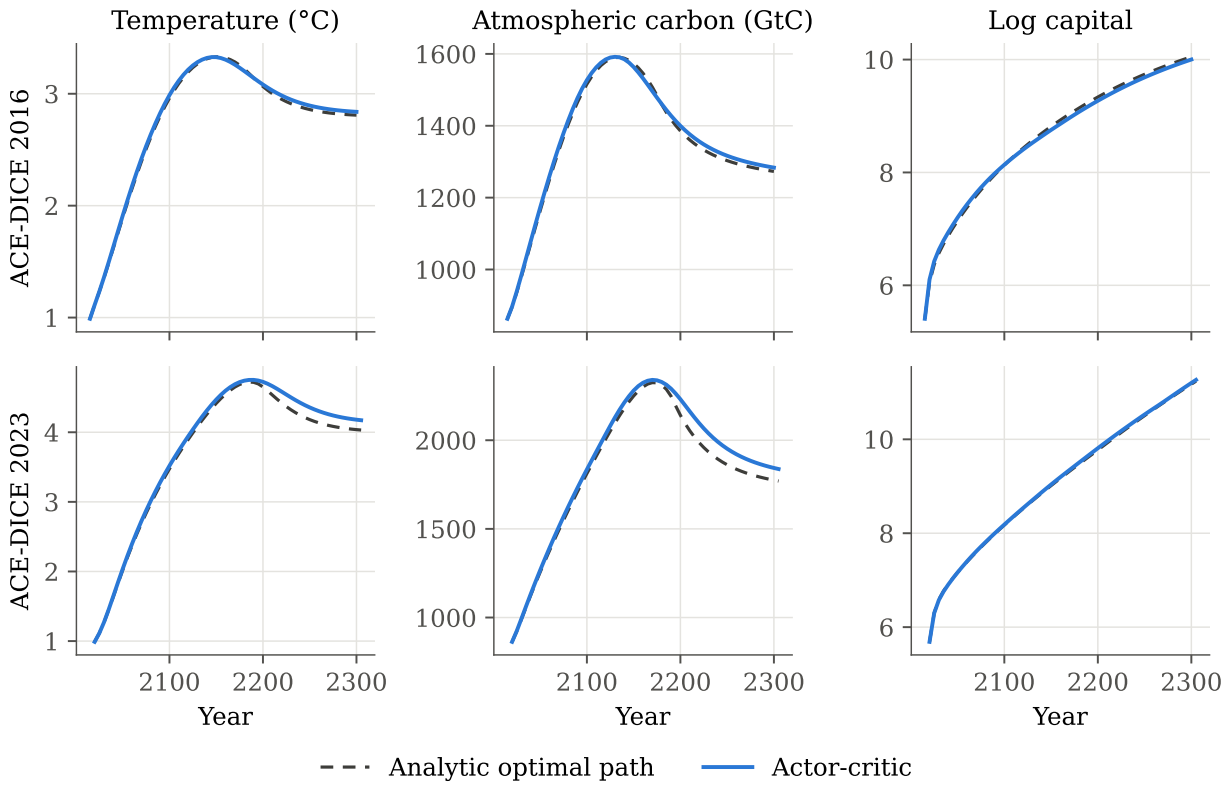


Figure 2: Temperature, atmospheric carbon and log capital under the learned policy and along the analytic optimal path from the same initial state.

and 36.9 USD/tCO$_2$ in the DICE 2023 model, against analytic values of 37.6 USD/tCO$_2$ in both.

6.3 Comparison with a Correctly Specified DEQN

The DEQN with correctly derived conditions is more precise than the actor-critic, by about an order of magnitude on the controls (Table 1). Its mean abatement error is 0.0009 and 0.0020, against 0.014 and 0.012 for the actor-critic, and its SCC errors are 0.2% and 0.6%, against 1.2% and 3.0%. This is the expected price of learning the shadow prices indirectly. The DEQN is told exactly which conditions the costates satisfy and outputs them directly, whereas the actor-critic must infer them from the curvature of a learned value function. On the other hand, the actor-critic obtains its accuracy without any hand-derived condition, so it cannot converge to the wrong solution because of a derivation error, and its gradient normalization replaces the hand-chosen scales of the DEQN. For a policy-relevant quantity such as the SCC, an error of 1–4% is well within the uncertainty of the underlying damage and climate parameters.

6.4 Learning Dynamics

Figure 5 shows how the errors evolve during training. During the warm-up, only the critic learns, and the implied carbon costate moves from a large, wrongly signed value towards the analytic one. At the end of the warm-up, it still has the wrong sign in both models (-0.14 and -0.65 times the

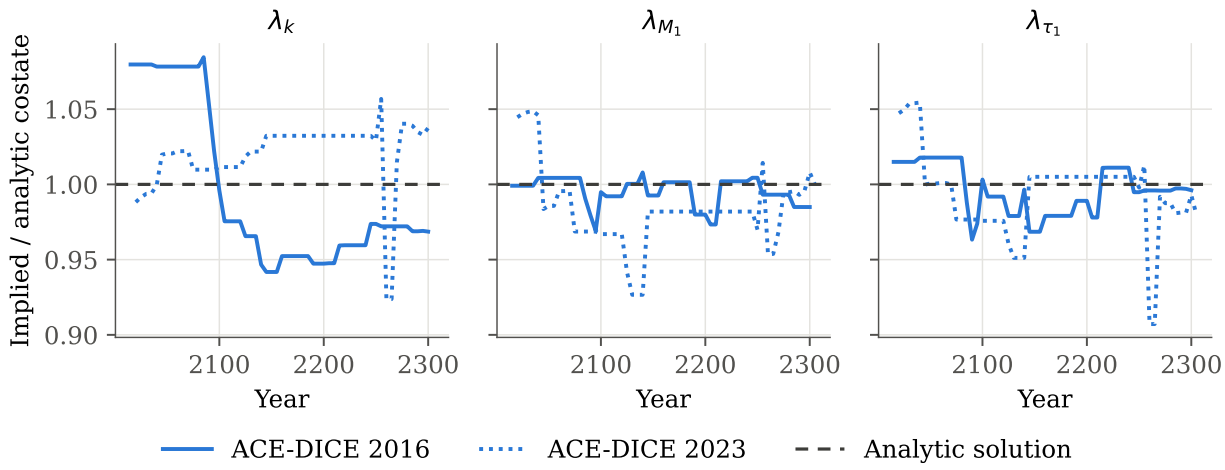


Figure 3: Costates implied by the critic, $\beta \nabla V_\phi(\boldsymbol{s}_{t+1})$, relative to their constant analytic values.

analytic value), and the actor briefly reduces abatement. The critic corrects the sign within 25 to 50 iterations, before the actor's abatement output saturates, and abatement then rises towards its optimal path. After 1000 iterations, the implied carbon costate fluctuates between 0.84 and 1.31 times its analytic value and settles at 0.98–1.00 as the learning rate decays, while the abatement error decreases steadily to about 0.013.

6.5 Ablations

Table 2 and Figure 6 report the two ablations for the DICE 2016 model. Both fail in the same way: the actor stops abating altogether.

Temporal-difference learning only. Without the value-gradient term, the critic fits the value of its policy *more* accurately than the full method, with a mean relative error of 0.02% against the true value of the policy. Its gradient with respect to carbon, however, has the wrong sign: the implied SCC in 2050 is -27.7 USD/tCO$_2$, so that emitting carbon appears to be beneficial, and the actor sets $\mu = 0$. The critic attributes part of the growth of the value over time, which is driven by technology and capital accumulation, to the carbon stock, which grows at the same time. This is the central lesson of the paper: an accurate value function is not sufficient for an accurate policy, because the policy depends on a gradient that is small relative to the value. The welfare loss of this policy is 0.40% of consumption, ten times that of the full method.

No state perturbation. Without perturbed states, the critic sees only the states along the trajectory, in which capital, carbon and temperature all move together. At the end of the warm-up, its carbon costate is still about -21 times the analytic value, and the actor drives abatement to zero within 50 iterations. The critic does eventually learn the carbon costate, to within 2.4% by the end of training, but by then the actor's abatement output is saturated at zero, where the gradient of the sigmoid vanishes, and the actor never recovers. The welfare loss is 0.16% of consumption.

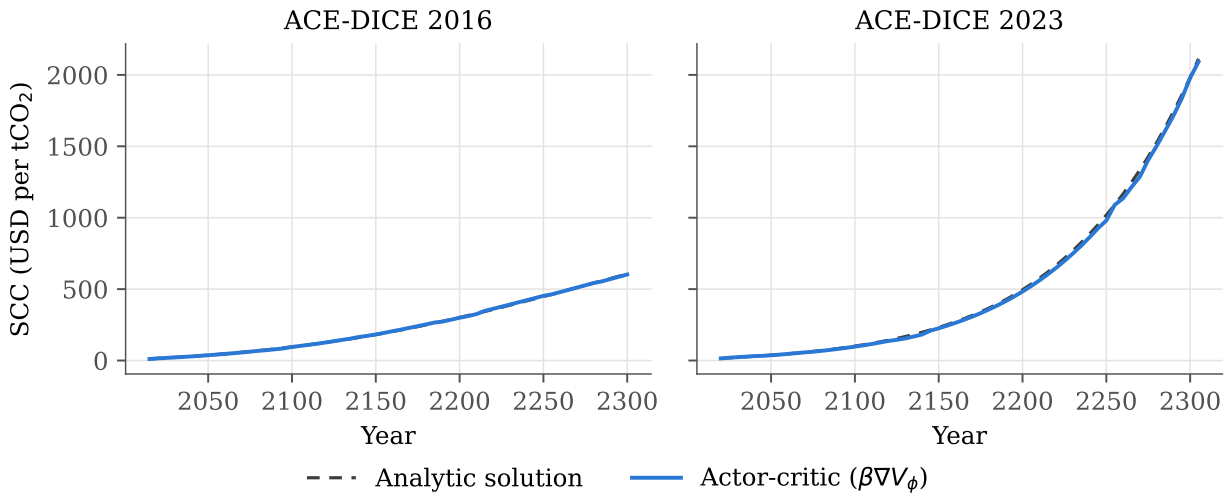


Figure 4: SCC implied by the critic and the analytic SCC, in USD per tCO_2.

	Full method	TD learning only	No state perturbation
$\max_t \lvert x_t - x^* \rvert$	0.0126	0.0716	0.0157
mean $\lvert \mu_t - \mu_t^* \rvert$	0.0141	0.6946	0.6940
$\max_t \lvert \mu_t - \mu_t^* \rvert$	0.0582	1.0000	0.9997
SCC, mean relative error	1.19%	187.40%	1.82%
λ_k, mean relative error	4.88%	17.65%	5.06%
λ_{M_1}, mean relative error	0.72%	191.99%	2.45%
λ_{τ_1}, mean relative error	1.42%	197.45%	1.18%
Welfare loss (consumption equivalent)	0.0388%	0.4007%	0.1623%

Table 2: Ablations for the DICE 2016 model. Errors are measured as in Table 1.

6.6 Robustness Across Seeds

The results above are for one random seed. Table 3 reports the same metrics for three seeds per model, which differ in the initialization of both networks and in the state perturbations. All six runs recover the analytic solution. The mean abatement error lies between 0.010 and 0.016, the largest deviation of the consumption rate between 0.007 and 0.014, the mean relative error of the SCC between 1.0% and 3.7%, and the welfare loss between 0.004% and 0.039% of consumption.

The learning dynamics differ more between seeds than the outcomes. In three of the six runs, the critic's carbon costate still had the wrong sign when the actor started, and the actor briefly reduced abatement to 2–3%. In each case, the critic corrected the sign within 100 to 300 iterations, before the actor's output saturated completely, and abatement recovered by iteration 800. The race described in Section 6.4 is therefore real, but with state perturbations, the critic wins it in all six runs. Without perturbations, it took the critic more than 1500 iterations to correct the sign, and the actor did not recover.

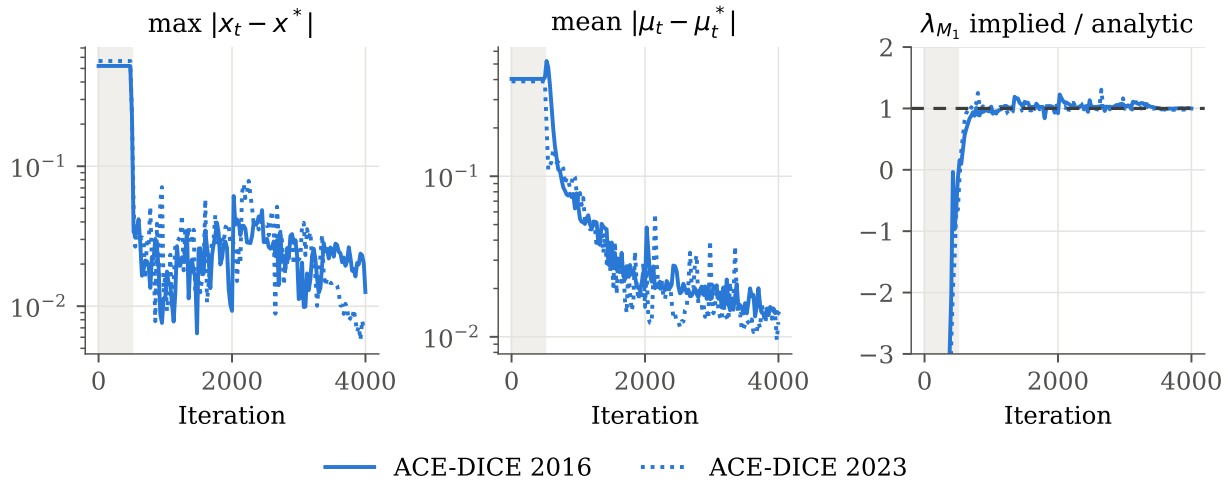


Figure 5: Errors of the learned policy and of the implied carbon costate during training, evaluated every 25 iterations. The shaded area is the critic warm-up.

Version	Seed	$\max_t \lvert x_t - x^* \rvert$	mean $\lvert \mu_t - \mu_t^* \rvert$	λ_k mean relative error	λ_{M_1}	SCC	Welfare loss (CE)
2016	1	0.0126	0.0141	4.88%	0.72%	1.19%	0.0388%
2016	2	0.0136	0.0150	2.02%	0.99%	1.01%	0.0374%
2016	3	0.0089	0.0160	2.84%	1.73%	1.80%	0.0200%
2023	1	0.0069	0.0123	2.66%	2.54%	2.99%	0.0039%
2023	2	0.0099	0.0104	1.88%	1.07%	1.04%	0.0157%
2023	3	0.0139	0.0137	4.99%	2.73%	3.70%	0.0160%

Table 3: Accuracy of the actor-critic across random seeds. Seed 1 is the run reported in the rest of the paper.

7 Discussion

Rewards and residuals. A residual-based loss and a reward are different objects. The reward is part of the economic model; a residual loss is a consistency criterion that a researcher derives from the model. The DEQN of the companion thesis illustrates the risk: with an envelope condition that omitted the direct effect of capital on utility and first-order conditions that counted the effect on future value twice, it converged, with a vanishing loss, to a consumption rate of 0.53–0.93 instead of 0.74. The actor-critic has no such failure mode. Its objective is built from the reward and the laws of motion, and the conditions that correspond to the envelope conditions, the gradient targets of the critic, are generated by automatic differentiation of the Bellman equation.

The SCC is a small gradient of a large function. The ablations show that the obstacle for a reward-based method is not the policy but the shadow price. A critic trained only on values can be very accurate and still imply the wrong sign for the SCC, because the marginal value of carbon is four orders of magnitude smaller than the value itself and is confounded with time along the

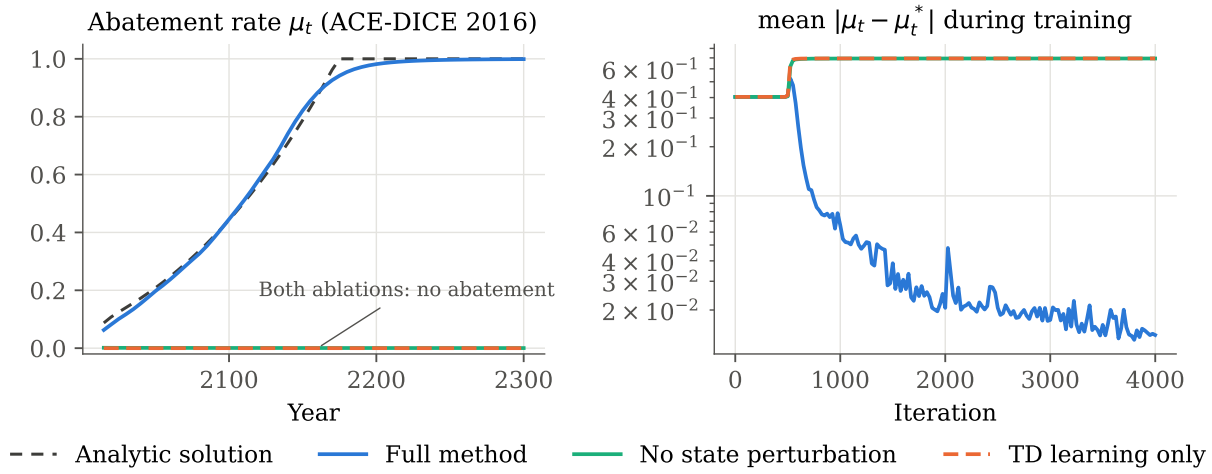


Figure 6: Abatement rate after training (left) and abatement error during training (right) for the full method and the two ablations, DICE 2016. Both ablations converge to zero abatement, so their lines overlap.

simulated trajectory. Both DEQNs and the actor-critic must therefore give the carbon conditions extra weight. In the DEQN, this requires hand-chosen scales for the costate outputs and residuals. In the actor-critic, it follows from normalizing each component of the gradient loss by the running scale of its own target, which requires no knowledge of the economics. Value-gradient learning is, in this sense, the natural critic for economic problems, in which policies are driven by shadow prices.

When residuals are still preferable. The correctly specified DEQN remains about an order of magnitude more precise on this problem, and residual-based methods have a decisive advantage beyond precision: they apply to equilibria that do not solve any single optimization problem. In decentralized economies with market clearing, heterogeneous agents or externalities that individual agents ignore, there is no single reward whose maximum is the equilibrium, but the equilibrium conditions can still be written as residuals. The reward-based approach is therefore a complement for problems with a planner, or with agents whose optimization problems can be learned separately, rather than a replacement.

Fragility. The method has failure modes of its own. The actor reacts to the critic's gradient within tens of iterations, while the critic's carbon gradient takes hundreds of iterations to converge. If the actor follows a wrongly signed critic long enough to saturate its output, it does not recover, even after the critic becomes correct (Section 6.5). The critic warm-up and the delayed actor updates address this race, but they are hyperparameters. I chose them in trial runs on the DICE 2016 model and applied them unchanged to the DICE 2023 model, and Section 6.6 shows that the outcome is robust to the random seed in six runs, although three of them briefly reduced abatement before recovering. Saturation could be avoided altogether by a parameterization of the actor that does not saturate, or by penalizing extreme outputs, which I leave for future work.

Limitations and extensions. The model is deterministic, the dynamics are known and differentiable, and business-as-usual emissions follow an exogenous path so that the benchmark is exact.

Each of these assumptions suggests an extension. First, with known differentiable dynamics, a critic is not strictly necessary: the policy can be trained by differentiating through simulated trajectories, with a learned value only at the end of a truncated horizon. Second, for stochastic or non-differentiable climate models, for instance with tipping points or with black-box emulators, model-free methods such as PPO or GRPO would become the relevant comparison. They estimate the policy gradient from sampled returns, and the small magnitude of the carbon signal suggests that they would need very large samples, which a comparison on this benchmark could quantify. Third, with endogenous business-as-usual emissions, the problem no longer has a closed form, but the actor-critic applies unchanged.

8 Conclusion

A model-based actor-critic, trained on the reward and the laws of motion alone, recovers the closed-form solution of the Analytic Climate Economy with DICE 2016 and DICE 2023 functional forms. Across six training runs, the learned abatement rate deviates from the optimal path by 0.010–0.016 on average, the consumption rate by at most 0.014, and the social cost of carbon read off the gradient of the critic is within 1–4% of its analytic value, with a welfare loss below 0.04% of consumption. A DEQN with correctly derived equilibrium conditions remains about an order of magnitude more precise, but the actor-critic needs no derived conditions and so cannot be misled by an error in deriving them.

The key ingredient is value-gradient learning. A critic trained on values alone fits the value function closely but implies a social cost of carbon with the wrong sign, and the actor stops abating. Training the critic on the gradient of the Bellman equation, computed by automatic differentiation and normalized by its own scale, recovers the shadow price of carbon. For economic applications, in which policy is driven by shadow prices that are small relative to the value function, this suggests that the gradient of the critic deserves as much attention as its level.

Code and Data Availability

All code, data and results are available at `https://github.com/hermanvest/DNN-for-ACE` in the directory `actor_critic/`.

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